**ANSWER KEY**

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · A1 · SKELETONS AND MEDICINES

# Bones and muscles

★☆☆
Grade 3

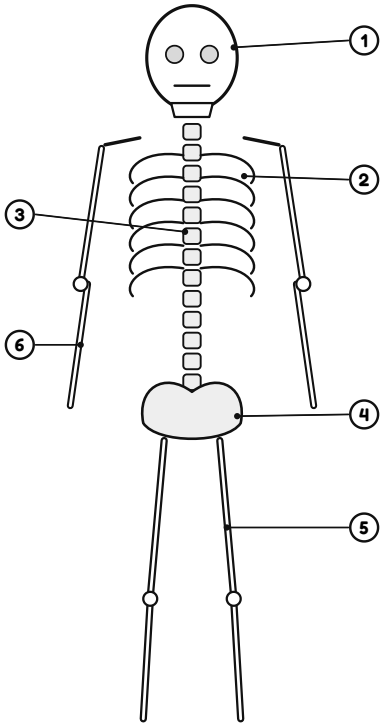
**Learning goals**

- Label the skeleton.
- Match bones to jobs.

## 1 Label the skeleton. [6]

LABELS

**spine (backbone)** **arm bones** **ribs** **pelvis** **skull** **thigh bone (femur)**




1 skull
2 ribs
3 spine (backbone)
4 pelvis
5 thigh bone (femur)
6 arm bones

## 2 Draw a line to match. [4]

Draw a line to match each pair, or write the letter.

| | | |
|---|---|---|
| skull ● | → A | ● **A** protects the brain |
| ribs ● | → C | ● **B** holds the body up |
| backbone ● | → B | ● **C** protect the heart and lungs |
| muscles ● | → D | ● **D** pull bones to move them |

## 3 True or false? [3]

| Statement | True | False | If false, correct it |
|---|---|---|---|
| 1. Muscles can push and pull. | | ✔ | Muscles can only pull. |
| 2. Adults have 206 bones. | ✔ | | |
| 3. It is safe to share medicine with a friend. | | ✔ | Never share medicine. |

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · A1 · SKELETONS AND MEDICINES

# How do we move?

★★☆
Grade 3

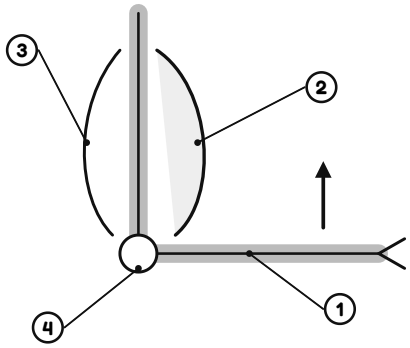
**Learning goals**

- Explain muscle pairs.

## 1 Label the arm muscles. [4]

LABELS

**triceps (relaxes)** **bone** **elbow joint** **biceps (contracts)**




1 bone

2 biceps (contracts)

3 triceps (relaxes)

4 elbow joint

## 2 What happens to the biceps when you bend your arm? [2]

It contracts (gets shorter and fatter) and pulls the lower arm up.

## 3 Why do muscles have to work in pairs? [2]

A muscle can only pull, so another muscle is needed to pull the bone back.

## 4 Why are some medicines kept in cupboards high up? [1]

So small children cannot reach them and take them by mistake.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · A1 · SKELETONS AND MEDICINES

# Do longer arms throw further?

★★☆
Grade 3

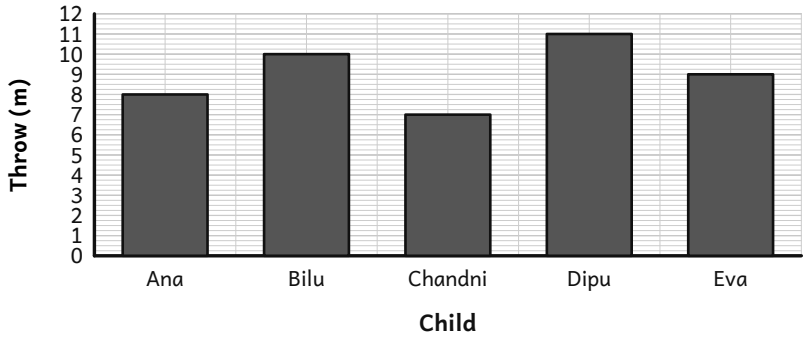
**Learning goals**

- Look for a pattern.
- Measure length.

Class 3B measured the arm length of five children and how far each could throw a bean bag (best of three throws).

**1** **Draw a bar chart.** [3]

| Child | Ana | Bilu | Chandni | Dipu | Eva |
|---|---|---|---|---|---|
| Arm length (cm) | 48 | 52 | 50 | 55 | 46 |
| Throw (m) | 8 | 10 | 7 | 11 | 9 |




**2** **Did the child with the longest arms throw furthest?** [1]

Yes: Dipu (55 cm) threw 11 m.

**3** **Is there a clear pattern? Explain.** [2]

Not always: Eva has the shortest arms but threw further than Chandni, so other things matter too (strength, practice).

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · A1 · SKELETONS AND MEDICINES

# Strong bones from small fish

★★★ Grade 3

**Learning goals**

- Link food to strong bones.
- Use numbers.

Science | Health | Food | Maths

Bones need **calcium** to grow strong. Milk, yoghurt (*doi*) and green leafy vegetables have calcium. In Bangladesh, tiny fish such as **mola** and **dhela** are eaten whole, with their bones, so they give lots of calcium too.

Scientists and nutrition workers in Bangladesh have encouraged farmers to grow mola fish in their ponds, so that children get more calcium and vitamins.

Based on nutrition research about small fish in Bangladesh.

**1** **Circle the foods that help build strong bones.** [3]




milk, small fish, green leaves

**2** **Why are small fish eaten whole so good for bones?** [2]

You eat the bones too, and fish bones contain calcium.

**3** **A child needs 3 glasses of milk a day. How many in a week?** [1]

3 × 7 = 21 glasses

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · A1 · SKELETONS AND MEDICINES

# Show what you know

★★★
Grade 3

**Learning goals**

- Use body words.

## 1 Find the words.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| R | E | P | M | U | S | C | L | E |
| I | Y | B | W | S | K | U | L | L |
| B | T | Y | M | H | U | S | L | B |
| S | P | H | I | M | B | O | W | O |
| M | E | D | I | C | I | N | E | N |
| D | Y | W | W | Y | N | D | D | E |
| A | B | U | N | V | S | J | S | D |
| A | H | N | R | O | S | U | G | L |
| R | E | K | Y | O | P | U | L | L |

**bone medicine muscle pull ribs skull**

## 2 Which bones protect the heart? [1]

○ skull
● **ribs**
○ leg bones

## 3 When a muscle contracts, it... [1]

○ gets longer
● **gets shorter and pulls**
○ pushes

## 4 Challenge. Animals like worms and jellyfish have no bones. How are their bodies different from ours? [2]

They are soft and bendy; they cannot stand up tall; they need water or soil to hold them.